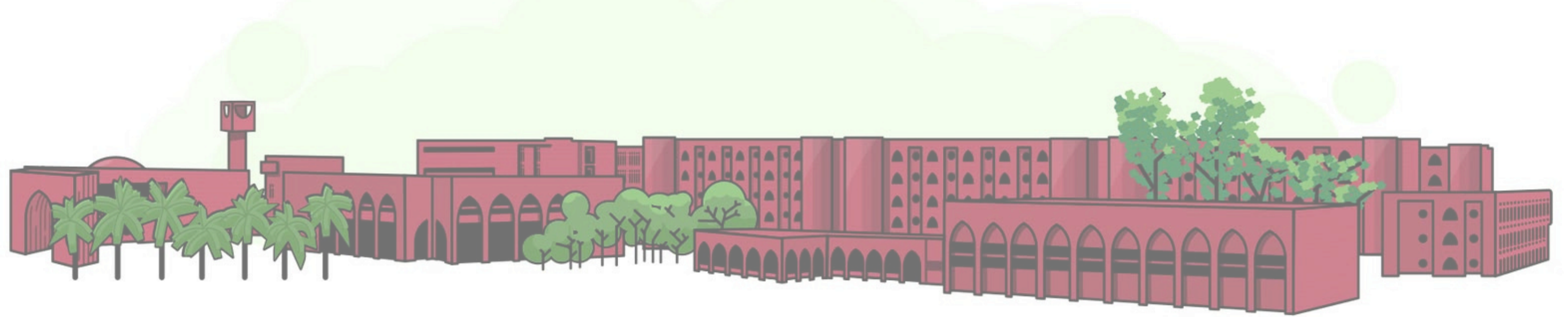




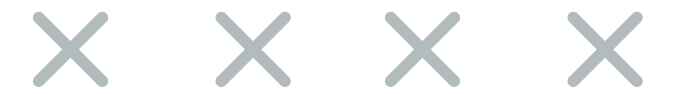
Team System322

IUTian Threads

Lab 9- Analyzing Systems Using Data Dictionaries

CSE 4408: System Analysis & Design Lab
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THE TEAM



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Project Recap



- IUTian Threads is a university-specific, forum-style platform designed for students to share concerns, ask questions, and engage in structured discussions.
- Addresses the current lack of a centralized and effectively moderated issue discussion system at IUT and IUT-OSW.

Data Flow Definition & Data Structure

Data Flow Definition

Data Flow Description		
ID : Name : Description :	DF02 New Post Contains all content and metadata when a user creates a new post in a subforum, including text, uploaded media and other related data.	
Source: Authenticated User (External Entity)		Destination: P2 - Post Interaction Handler
Type of the Data Flow: ☐ File ☐ Screen ☐ Report ☒ Form ☐ Internal		
Associated Data Structure: New_Post	Volume/Time: ~60/day (expected estimates at launch)	Comments: Must be submitted by an authenticated user; attachments scanned for prohibited content; stored with timestamp for moderation/audit.

Data Structure Definition

Chosen data structure: New_Post

New_Post = Post_Title + User_ID + Post_Content + Post_Timestamp + Media_Attachments + Post_Type

Post_Title = String

User_ID = {String | Null} (Null for anonymous post, authenticated by session cookie)

Post_Content = String

Post_Timestamp = DateTime

Media_Attachments = {Image | Video | None}

Post_Type = {Anonymous | Normal}



Data Elements & Data Store

Data Element Definition-1

ID	: DE01	
Name	: Post_Title	
Alias	: Title	
Description	: The short, descriptive title for a post, helping users understand the content of the post at a glance	
Element Characteristics		
Length	: Variable (max 200 character)	Data Type: String
Format	: String (UTF-8 encoded)	
Default Value	: None	
Type	: ☒ Base ☐ Derived	
Validation Criteria: Cannot be NULL; Must contain at least one alphanumeric character; special characters allowed		
Comments: Stored as a String in a NoSQL document, after input sanitization allowing for flexibility with content length; Supports case-insensitive search for user queries		

Data Element Definition-2

ID	: DE02	
Name	: User_ID	
Alias	: UID	
Description	: A unique identifier for each student within the system. ID is assigned to track student’s actions.	
Element Characteristics		
Length	: 36 characters (UUID format)	Data Type: String
Format	: UUID (e.g., 550e8400-e29b-41d4-a716-446655440000)	
Default Value	: None	
Type	: ☒ Base ☐ Derived	
Validation Criteria: Must be a valid UUID (or a reference ID in NoSQL); Must exist in User Master collection		
Comments: Used across the system to link users with their posts, votes, and comments.		

Data Element Definition-3

ID	: DE03		
Name	: Post_Content		
Alias	: Body		
Description	: The main contents of the post		
Element Characteristics			
Length	: Variable (Max 10,000 characters, unstructured)	Data Type:	String or Text (NoSQL text fields)
Format	: Free text, HTML allowed (Rich content support)		
Default Value	: None		
Type	: ☒ Base ☐ Derived		
Validation Criteria: Cannot be NULL; No specific character length limit beyond max (10000);			
Comments: Supports Markdown structure also; content is subject to filtering for inappropriate material			

Data Store Definition

ID Name Alias Description	: D1 : In-subforum Content Master : Content_Master : Stores all posts, comments, and metadata within each subforum, including post content, media attachments, and user interactions.
Data Store Characteristics	
File Type Format	: Binary : Document Store (e.g. MongoDB)
Associated Data Structure: New_Post (document) Primary Key: Post_ID (Auto-generated, unique) Secondary Keys: User_ID, Subforum_ID, Post_Timestamp	
Comments: Indexed by Post_Timestamp and Subforum_ID for quick sorting and retrieval; Supports media files (image/video) as embedded or linked documents	



Ambiguities

Needed to decide the limit for post body between a smaller limit for performance (e.g., 5,000 chars) or a larger limit for user flexibility (e.g., 10,000 chars).

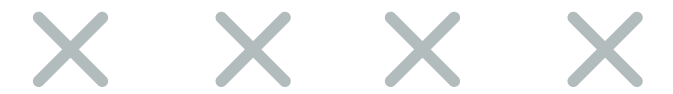
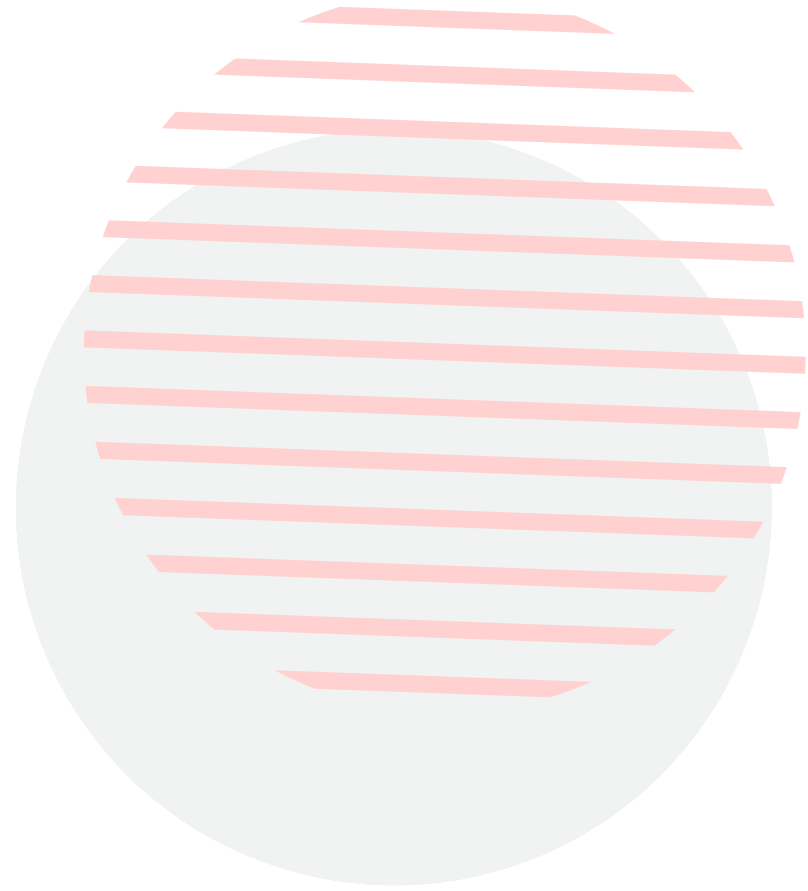
Challenges

Ensuring no malicious code is embedded in text or attachments while allowing user to use styled text using Unicode characters



Discoveries

Storing attachments in external storage and referencing them with URLs drastically reduces DB size (BLOBs) and improves query speed



Thank you!

We are open for Q&A

